Question 1 – Critique and Redesign an ECDF and Q-Q Plot

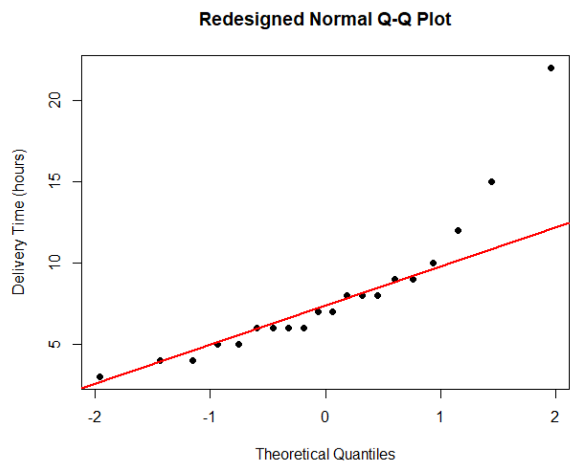
```
data <- data.frame(
  Order_ID = paste0("O", sprintf("%02d", 1:20)),
  Warehouse = c(
    "Chennai","Chennai","Chennai","Chennai","Chennai",
    "Bengaluru","Bengaluru","Bengaluru","Bengaluru","Bengaluru",
    "Hyderabad","Hyderabad","Hyderabad","Hyderabad","Hyderabad",
    "Pune","Pune","Pune","Pune","Pune"
  ),
  Delivery_Time_Hrs = c(
    4,5,6,7,9,
    3,5,6,8,10,
    6,7,8,12,15,
    4,6,8,9,22
  )
)

print(data)
plot(
  ecdf(data$Delivery_Time_Hrs),
  main = "Original ECDF - Delivery Time",
  xlab = "Delivery Time",
  ylab = "ECDF",
  col = "blue",
  lwd = 2,
  xlim = c(0,25),
  ylim = c(0,1),
  las = 1
)
```

```
grid()
qqnorm(
  data$Delivery_Time_Hrs,
  main = "Original Q-Q Plot - Delivery Time",
  xlab = "Theoretical Quantiles",
  ylab = "Sample Quantiles"
)
grid()
# Notice: No qqline() is added here intentionally.
# This represents the problematic original-style plot
cat("\nVISUALIZATION PROBLEMS:\n")
  lwd = 3,
  xlim = c(0,25),
  ylim = c(0,1),
  yaxt = "n"
)
axis(
  2,
  at = seq(0,1,0.1),
  labels = paste0(seq(0,100,10), "%")
)
abline(
  h = seq(0.1,0.9,0.1),
  col = "gray90",
  lty = 3
)
# Highlight extreme value
points(
  22,
  ecdf(data$Delivery_Time_Hrs)(22),
```

```
  pch = 19,
  col = "red"
)
  lwd = 2
)
cat("\nSUMMARY STATISTICS:\n")
cat("Mean:",
    mean(data$Delivery_Time_Hrs), "hours\n")
cat("Median:",
    median(data$Delivery_Time_Hrs), "hours\n")

cat("Standard Deviation:",
    sd(data$Delivery_Time_Hrs), "hours\n")
cat("Maximum:",
    max(data$Delivery_Time_Hrs), "hours\n")
cat("Minimum:",
    min(data$Delivery_Time_Hrs), "hours\n")
Q1 <- quantile(data$Delivery_Time_Hrs, 0.25)
Q3 <- quantile(data$Delivery_Time_Hrs, 0.75)
IQR_value <- Q3 - Q1
lower_limit <- Q1 - 1.5 * IQR_value
upper_limit <- Q3 + 1.5 * IQR_value
outliers <- data$Delivery_Time_Hrs[
  data$Delivery_Time_Hrs < lower_limit |
  data$Delivery_Time_Hrs > upper_limit
]
```

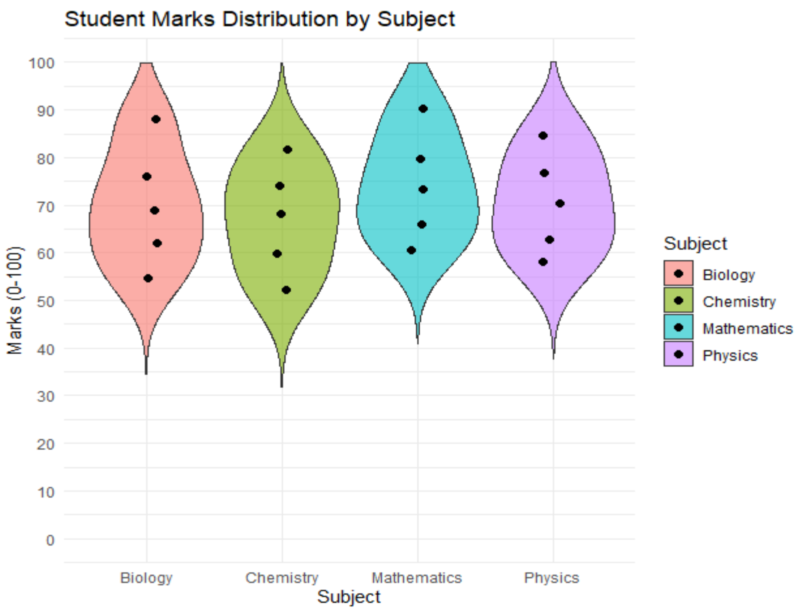


Question 2 – Critique and Redesign Many Distributions Using Bean Plots

```
# Four separate boxplots with different y-axis ranges
par(mfrow = c(2,2))
boxlot(
  data$Marks[data$Subject == "Physics"],
  main = "Physics",
  ylab = "Marks",
  ylim = c(50,90)
)
boxplot(
  data$Marks[data$Subject == "Chemistry"],
  main = "Chemistry",
  ylab = "Marks",
  ylim = c(45,85)
)
boxplot(
  data$Marks[data$Subject == "Mathematics"],
  main = "Mathematics",
  ylab = "Marks",
  ylim = c(55,95)
)
boxplot(
  data$Marks[data$Subject == "Biology"],
  main = "Biology",
  ylab = "Marks",
  ylim = c(50,90)
)
par(mfrow = c(1,1))
  labs(
    title = "Distribution of Marks Across Subjects",
```

```
    x = "Subject",
    y = "Marks"
  ) +
  ) +
  theme_minimal(
summary_data <- aggregate(
  Marks ~ Subject,
  data = data,
  FUN = function(x) c(
    Mean = mean(x),
    Median = median(x),
    SD = sd(x),
    Minimum = min(x),
    Maximum = max(x),
    Range = max(x) - min(x)
  )
)
print(summary_data)
# Calculate range for each subject
range_data <- aggregate(
  Marks ~ Subject,
  data = data,
  FUN = function(x) max(x) - min(x)
)
range_data <- range_data[
  order(-range_data$Marks),
]
cat("\nSUBJECTS RANKED BY SPREAD:\n")
print(range_data
# Find subject with greatest spread
```

```
greatest_spread <- range_data$Subject[1]
cat(
  "\nSubject with greatest spread:",
  greatest_spread,
  "\n"
)
```

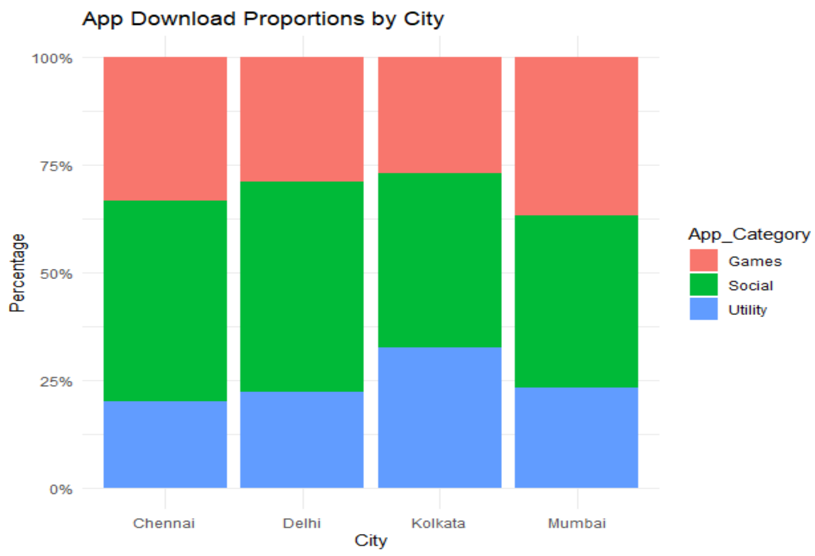


Question 3 – Critique and Redesign Proportion Visualizations

```
# Separate data for each city
chennai <- data[data$City == "Chennai", ]
mumbai <- data[data$City == "Mumbai", ]
delhi <- data[data$City == "Delhi", ]
kolkata <- data[data$City == "Kolkata", ]
# Create 4 pie charts
par(mfrow = c(2,2))
pie(
  chennai$Downloads,
  labels = chennai$App_Category,
  main = "Chennai",
  explode = c(0.1,0,0)
)
pie(
  mumbai$Downloads,
```

```
  labels = mumbai$App_Category,
  main = "Mumbai",
  explode = c(0,0.1,0)
)
pie(
  delhi$Downloads,
  labels = delhi$App_Category,
  main = "Delhi",
  explode = c(0,0.1,0)
)
pie(
  kolkata$Downloads,
  labels = kolkata$App_Category,
  main = "Kolkata",
  explode = c(0,0,0.1)
)
par(mfrow = c(1,1))
ggplot(
  data,
  aes(
    x = City,
    y = Downloads,
    fill = App_Category
  )
) +
  geom_bar(
    stat = "identity",
    position = "dodge"
  ) +
  labs(
```

```
    title = "App Downloads by City and Category",
    x = "City",
    y = "Downloads"
  ) +
  theme_minimal()
ggplot(
  data,
  aes(
    x = City,
    y = Downloads,
    fill = App_Category
  )
) +
  geom_bar(
    stat = "identity",
    position = "fill"
  ) +
  scale_y_continuous(
    labels = function(x) paste0(x * 100, "%")
  ) +
  labs(
    title = "App Download Proportions by City",
    x = "City",
    y = "Percentage"
  ) +
  theme_minimal()
cat("\nCOMPARISON:\n")
```



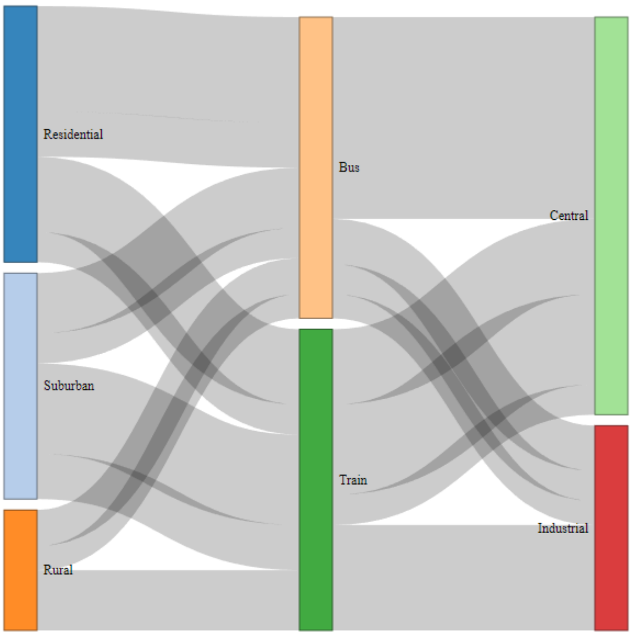
Question 4 – Critique and Redesign Nested Proportions

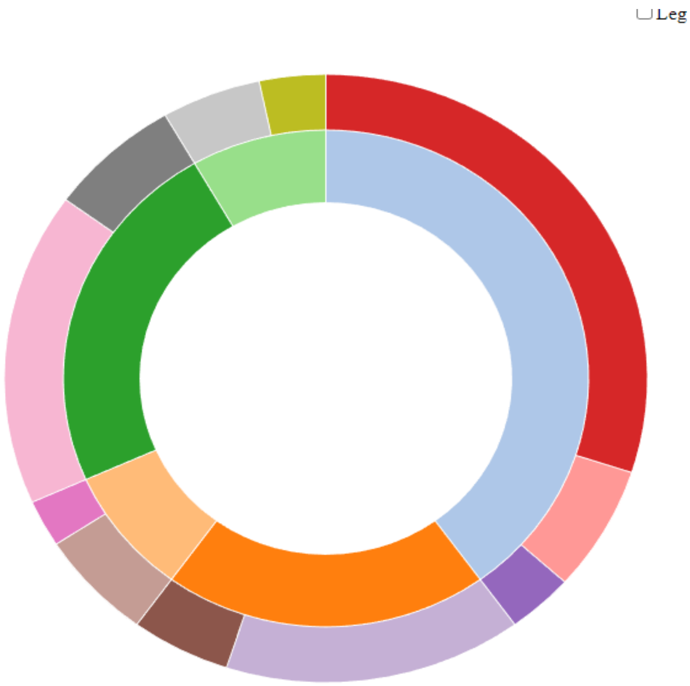
```
treemap(
  data,
  index = c("Campus", "Department", "Expense_Type"),
  vSize = "Amount",
  title = "Original Campus Expenditure Treemap"
)
cat("\nCRITIQUE OF ORIGINAL TREEMAP:\n")


cat("1. Small areas make some expense labels difficult to read.\n")
cat("2. The color scale does not clearly distinguish the hierarchy.\n")
cat("3. Different hierarchy levels are not visually separated clearly.\n")
cat("4. Similar-sized rectangles can be difficult to compare.\n")
cat("5. Too many labels can create visual clutter.\n")
treemap(
  data,
  index = c("Campus", "Department", "Expense_Type"),
  vSize = "Amount",
  vColor = "Amount",
  type = "value",
  palette = "Blues",
  border.col = "white",
  border.lwds = c(3, 2, 1),
```

```r
  fontsize.labels = c(14, 11, 9),
  fontcolor.labels = c("black", "black", "black"),
  align.labels = list(
    c("left", "top"),
    c("left", "top"),
    c("center", "center")
  ),
  title = "Improved Campus Expenditure Treemap"
)
sunburst_data <- data.frame(
  sequence = paste(
    data$Campus,
    data$Department,
    data$Expense_Type,
    sep = "-"
  ),
  size = data$Amount
)
sunburst(
  sunburst_data,
  count = TRUE
)
cat("\nTREEMAP VS SUNBURST:\n")
cat("\nTreemap:\n")
cat("Uses rectangle area to show expenditure size.\n")
cat("Makes large spending categories easy to identify.\n")
cat("However, small areas may have difficult-to-read labels.\n")
cat("\nSunburst:\n")
cat("Uses concentric rings to show Campus -> Department -> Expense Type.\n")
cat("Makes the hierarchy very clear.\n")
```

cat("However, comparing similar-sized segments can be more difficult.\n")





Question 5 – Critique and Redesign Flow-Based Proportions

```
nodes <- data.frame(
  name = c(
    unique(data$Course),
    unique(data$Certification),
    unique(data$Job_Role)
  )
)


# Course -> Certification
link1 <- aggregate(
  Students ~ Course + Certification,
  data = data,
  sum
)
# Certification -> Job Role
link2 <- aggregate(
```

```
  Students ~ Certification + Job_Role,
  data = data,
  sum
)
links <- data.frame(
  source = c(
    match(link1$Course, nodes$name) - 1,
    match(link2$Certification, nodes$name) - 1
  ),
  target = c(
    match(link1$Certification, nodes$name) - 1,
    match(link2$Job_Role, nodes$name) - 1
  ),
  Value = "value",
  NodeID = "name",
  fontSize = 13,
  nodeWidth = 35,
  sinksRight = TRUE
)
# Add descriptive labels to nodes
nodes_redesign <- data.frame(
  name = c(
    "Course: Data Science",
    "Course: Web Development",
    "Course: Cloud Computing",
    "Course: UI/UX Design",
    "Certification: Yes",
    "Certification: No",
    "Job: Analyst",
    "Job: Engineer",
```

```
    "Job: Frontend Dev",
    "Job: Backend Dev",
    "Job: DevOps Engineer",
    "Job: Cloud Analyst",
    "Job: Support Engineer",
    "Job: Product Designer",
    "Job: UX Researcher"
  )
)
# Create labelled links manually
# Course -> Certification
source_redesign <- c(
value_redesign <- c(
  36,0,9,
  30,0,7,
  27,0,6,
  21,0,5
)
keep <- value_redesign > 0
links_course <- data.frame(
  source = source_redesign[keep],
  target = target_redesign[keep],
  value = value_redesign[keep]
)

# Certification -> Job Role
links_job <- data.frame(
  source = c(
    4,4,5,
    4,4,5,
```

```
    4,4,5,
    4,4,5
  ),
redesign_links <- rbind(
  links_course,
  links_job
)
sankeyNetwork(
  Links = redesign_links,
  Nodes = nodes_redesign,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 13,
  nodeWidth = 40,
  sinksRight = TRUE
)
# Create an ID for each flow
data$Flow_ID <- 1:nrow(data)
# Convert data to long format
parallel_data <- data.frame(
  Flow_ID = rep(data$Flow_ID, 3),
  Stage = rep(
    c("Course", "Certification", "Job Role"),
    each = nrow(data)
  ),
  Category = c(
    data$Course,
    data$Certification,
```

```
    data$Job_Role
  ),
  Students = rep(data$Students, 3)
print(largest)
cat("\nFLOW DETAILS:\n")
for(i in 1:nrow(largest)) {
  cat(
    largest$Course[i],
    "-> Certification:", largest$Certification[i],
    "->", largest$Job_Role[i],
    ":",
    largest$Students[i],
    "students (",
    round(largest$Percentage[i], 2),
    "%)\n"
  )
}
```